

Lloyd-Dexter Akoulouze Bika

+1 (210) 275-3201 | dexter5506@gmail.com | [linkedin.com/in/lloyd-dexter-akoulouze-bika](https://www.linkedin.com/in/lloyd-dexter-akoulouze-bika) | <https://dext3r.azurewebsites.net/>

OBJECTIVE

Driven to excel by putting forth time and effort most won't, I am a passionate learner with a deep interest in machine learning, mathematics, and software engineering. I have conducted several projects independently, demonstrating my proactive nature and resourcefulness. I aim to leverage my strong work ethic and self-driven approach to contribute to the future of technology by continuing to grow both personally and professionally.

EDUCATION

The University of Texas at San Antonio

MDS in Artificial Intelligence (In progress)

San Antonio, TX

Expected Graduation: May 2028

Tom C. Clark High School

High School Diploma — GPA: 4.0/4.0

San Antonio, TX

Aug 2020 – June 2024

PROJECTS

Reinforcement Learning Snake AIs | *Python, Deep Q-Learning, Pygame*

Aug 2023 – Sep 2023

- Developed a Snake game simulation using Pygame, creating an interactive canvas for game play.
- Implemented an exploration-exploitation strategy to optimize the agent's learning process.
- Utilized Deep Q-Learning to train the AI agent, applying and iteratively refining a reward system that enhanced decision-making and improved the efficiency of the training process.
- Gained hands-on experience in developing, testing, and refining reward systems, which is crucial for real-world AI applications like autonomous navigation and decision-making systems.

Lidar Detection and SLAM Simulation | *Python, Pygame*

June 2023 – July 2023

- Simulated a 2D Lidar sensor within a Pygame environment to replicate real-world obstacle detection capabilities, including the dynamic visualization of sensor data.
- Developed a system that accurately identifies black regions in an image (e.g., house layout) and overlays red points on the Pygame canvas to simulate the sensor's detection of obstacles.
- Gained practical experience in implementing and testing SLAM (Simultaneous Localization and Mapping) techniques, deepening understanding of their applications in autonomous navigation systems.

MNIST Handwritten Digit Classification | *Python, Keras, MNIST*

June 2024 – Aug 2024

- Built and trained a neural network model for handwritten digit classification using the MNIST dataset, leveraging the Keras library to construct and fine-tune the model.
- Optimized the model by experimenting with different activation functions (e.g., ReLU, softmax), number of layers, and neurons per layer, achieving 98% classification accuracy.
- Addressed challenges like overfitting using techniques such as dropout and early stopping, ensuring the model maintained high performance on unseen data.
- Adjusted the network architecture to improve training efficiency, refining hyperparameters like learning rate and batch size.
- Enhanced understanding of machine learning workflows such as backpropagation and gradient descent, which are critical in real-world AI applications like optical character recognition (OCR) and autonomous vehicle systems.

RELEVANT EXPERIENCE

Arduino & Raspberry Pi

- learned to interface hardware with software, manage GPIO pins, and work with protocols like I2C or SPI
- Programmed systems to collect and process data or trigger actions based on inputs, including sensors and actuators for automation tasks.
- Designed circuits and managed power distribution for small-scale projects

Solidworks

- Gained proficiency in 3D modeling, sketching, and assembling parts, applying these skills in a robotic arm project to design and simulate movements.
- Developed an understanding of dimensioning, tolerances, and constraints, which are crucial in mechanical design

TECHNICAL SKILLS

Languages: Python, Bash, SQL, HTML, CSS, JavaScript

Frameworks and Libraries: Django, Keras, Pygame

Tools and Platforms: GitHub, Azure, SolidWorks, Raspberry Pi

Design Software: Adobe Premiere Pro